

Flutter Material Design & Material 3 Report

This report covers the implementation and understanding of Flutter Material Design widgets and Material 3 components. The project focuses on building beautiful, responsive, and user-friendly interfaces using Flutter widgets such as buttons, cards, dialogs, forms, SnackBars, navigation systems, and validation techniques.

Learning Objectives

- Master Material Design widgets
- Implement Material 3 components
- Use buttons, cards, dialogs, and forms
- Build responsive and beautiful Flutter UIs

Material Design Widgets

ElevatedButton: Used for important actions with elevation and background color.

TextButton: A flat button used for less important actions.

OutlinedButton: A button with an outlined border.

IconButton: A clickable icon button.

FloatingActionButton: Circular floating button for primary actions.

Cards and ListTile

The Card widget is used to create modern UI cards with elevation and rounded corners. ListTile simplifies creating structured lists with leading icons, titles, subtitles, and trailing widgets.

- Important Components:
 - Card widget
 - ListTile
 - Leading widgets
 - Trailing widgets
 - Divider widget

Dialogs and Bottom Sheets

- AlertDialog – Used for confirmation and alerts.
- SimpleDialog – Displays multiple options.
- showDialog() – Displays dialogs on screen.
- Custom Dialogs – Fully customizable dialogs.

- BottomSheet – Slides from the bottom of the screen.

SnackBar and Material Banner

SnackBars provide temporary notifications at the bottom of the screen. Material Banners display persistent messages with actions.

- Showing SnackBars using ScaffoldMessenger
- Adding SnackBar actions
- Displaying MaterialBanner
- Handling user interactions

Forms and Validation

Flutter forms are used to collect user input and validate data before submission.

- TextField
- TextFormField
- Form widget
- Validation
- InputDecoration

Navigation in Flutter

- Navigator.push() – Opens a new screen.
- Navigator.pop() – Returns to previous screen.
- Named Routes – Simplifies route management.
- Passing data between screens.

Material Design Showcase App

A complete Flutter application was developed to demonstrate Material Design concepts.

- Login and Register Screens
- Dashboard with Cards
- Forms with Validation
- Dialogs and SnackBars
- Navigation between multiple screens
- Material 3 UI Components

Deliverables

- Material Design showcase app
- 10+ Material widgets used
- Navigation between screens

- Forms with validation
- Dialogs and SnackBar working
- README with Material Design concepts

Project Structure

```
lib/  
  main.dart          # app entry and route definitions  
  screens/           # per-screen UI pages  
    home_screen.dart  
    buttons_screen.dart  
    cards_screen.dart  
    forms_screen.dart  
    dialogs_screen.dart
```

Conclusion

This project provided hands-on experience with Flutter Material Design and Material 3 components. The implementation improved understanding of UI development, navigation, forms, dialogs, and responsive design techniques in Flutter.